

03

SECTION TITLE

2.3

Electron Configurations

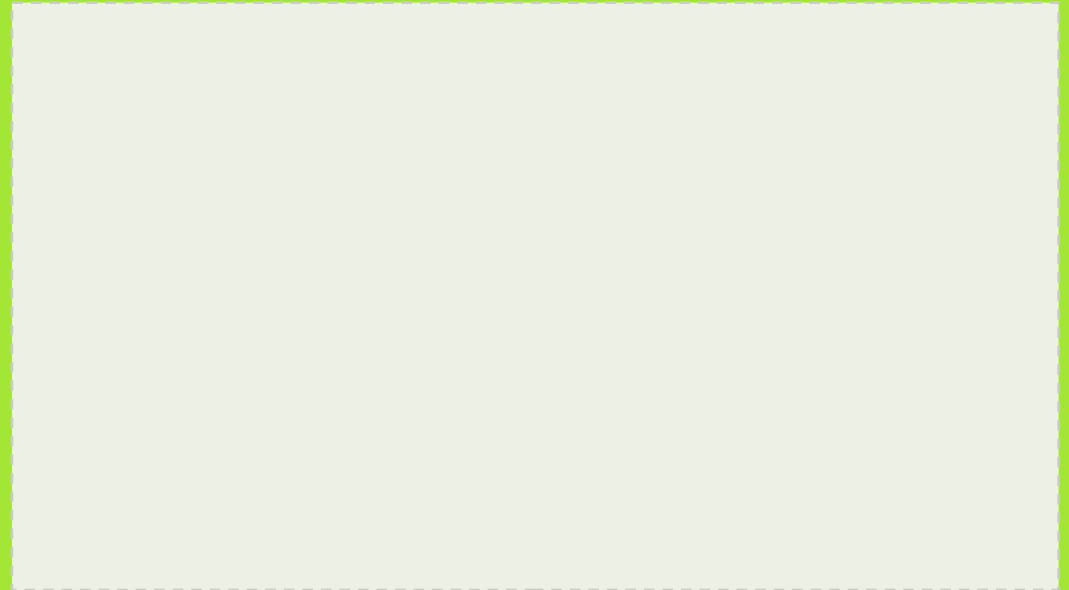
Write configurations and orbital diagrams through krypton, and catch mistakes in them.

Where Are Oxygen's Electrons?

The periodic table gives you the count: 8. Today you write where each one sits.

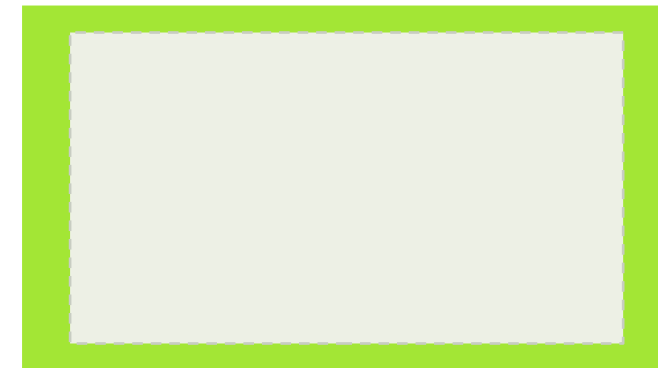
KEY QUESTION

Oxygen has 8 electrons. Do all 8 have the same energy?



Reading an Electron Configuration

Take $2p^4$ apart one piece at a time.



ENERGY LEVEL

$2p^4$

The big number in front is the level: 2.

SUBLEVEL

$2p^4$

The letter is the sublevel: p.

ELECTRONS

$2p^4$

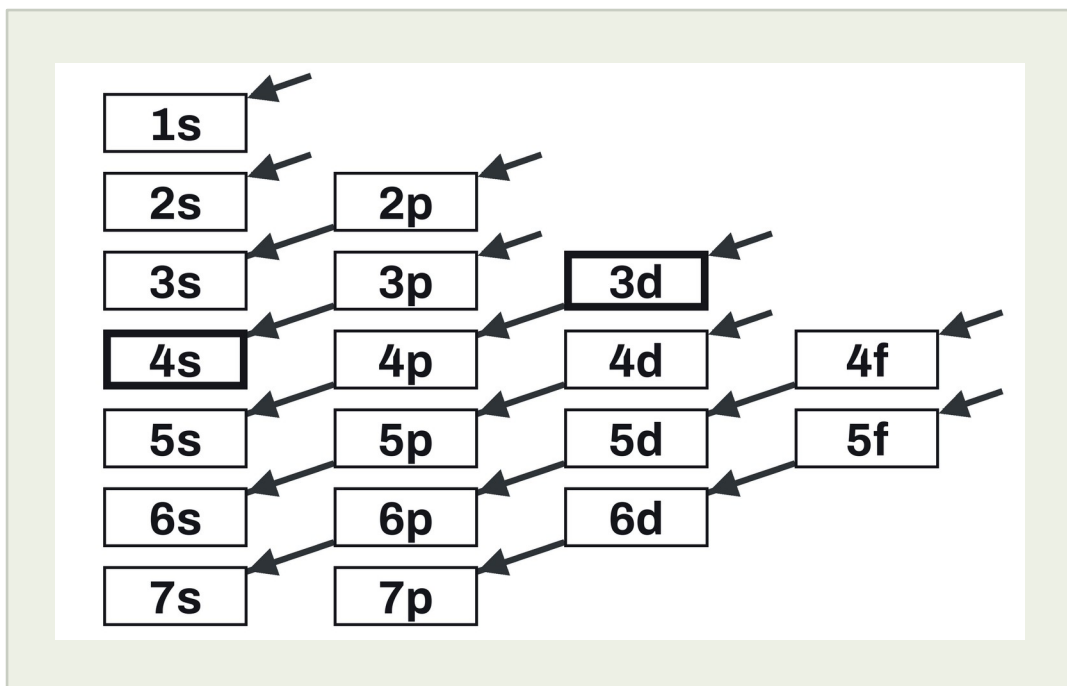
The raised number is the electron count: 4.

MUST WRITE

A configuration lists where the electrons sit. Ground state means the lowest-energy arrangement.

The Order Electrons Fill In

Read each arrow from top right to bottom left, then start the next arrow.



AUFBAU PRINCIPLE

The lowest-energy open orbital fills first.

THE ORDER

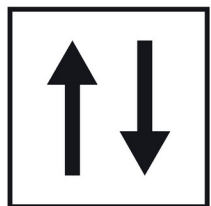
1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p

HEAVY BOXES

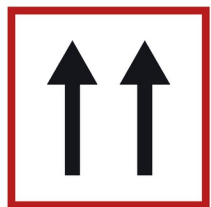
4s fills before 3d, in K and Ca.

Pauli: Two per Orbital, Opposite Spins

Up and down arrows are spin labels, not electrons orbiting opposite ways.



✓ **CORRECT**
opposite spins



✗ **WRONG**
same spin

PAULI EXCLUSION PRINCIPLE

One orbital holds at most 2 electrons.

THE SPINS

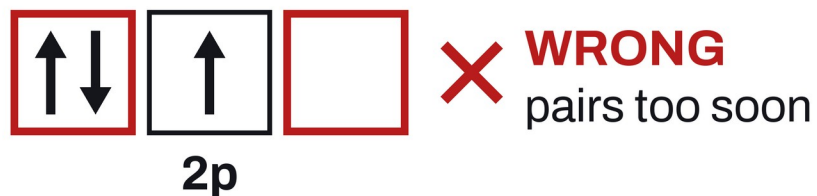
If there are two, their spins are opposite.

COMMON MIX-UP

Two per orbital, not two per sublevel.

Hund's Rule: Singles Before Pairs

Equal-energy orbitals, like the three 2p boxes, each get one electron first.



HUND'S RULE

Singles first, all pointing the same way.

NITROGEN

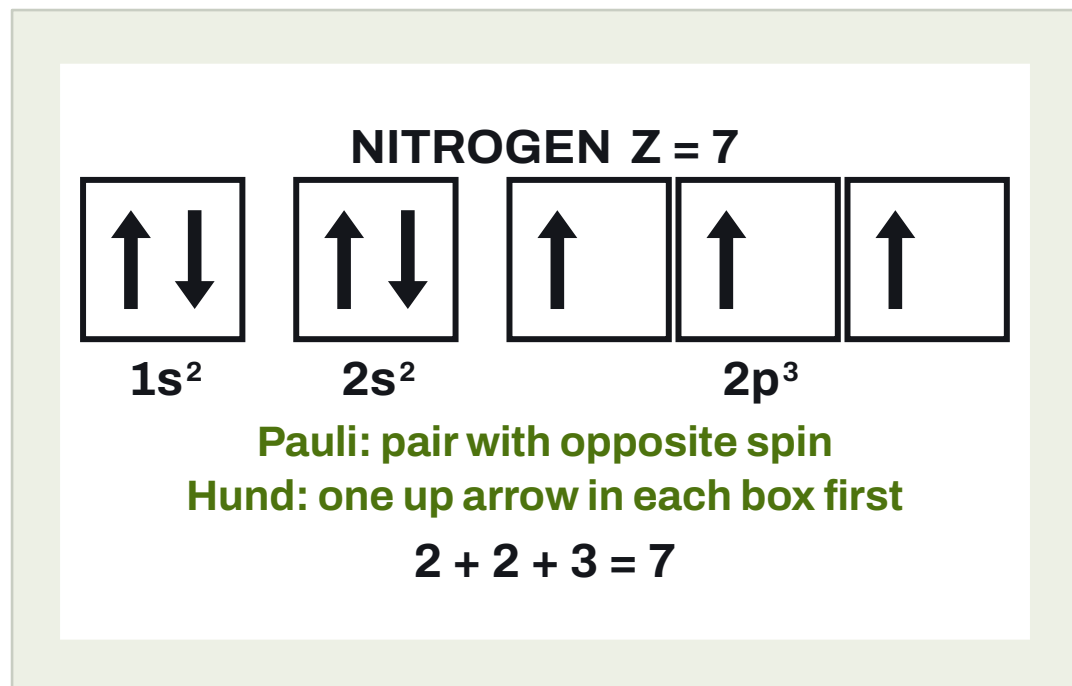
Its $2p^3$ is three single electrons: 3 unpaired.

THE TERM

Equal-energy orbitals are called degenerate.

Reading an Orbital Diagram

One box per orbital. Nitrogen has 7 electrons, so 7 arrows.



BOXES

1s and 2s have 1 box each. 2p has 3.

ARROWS

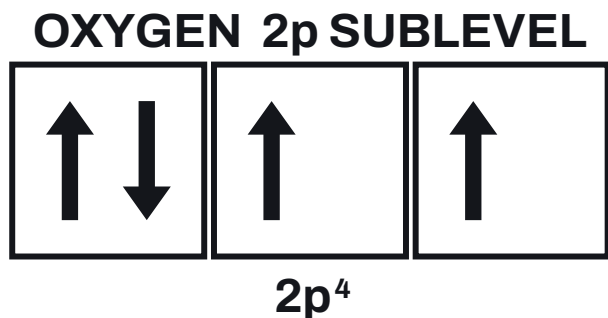
Draw the up arrow first in every box.

THE LABEL UNDER EACH GROUP

The raised number counts the arrows.

Filling Oxygen's 2p Boxes

Four electrons, three boxes. Watch the order of the arrows.



Hund: one up arrow in each box first

Pauli: pair with opposite spin

4 arrows = 4 electrons in 2p

THE BOXES

The 2p sublevel has 3 boxes. Oxygen puts 4 electrons in them.

HUND'S RULE

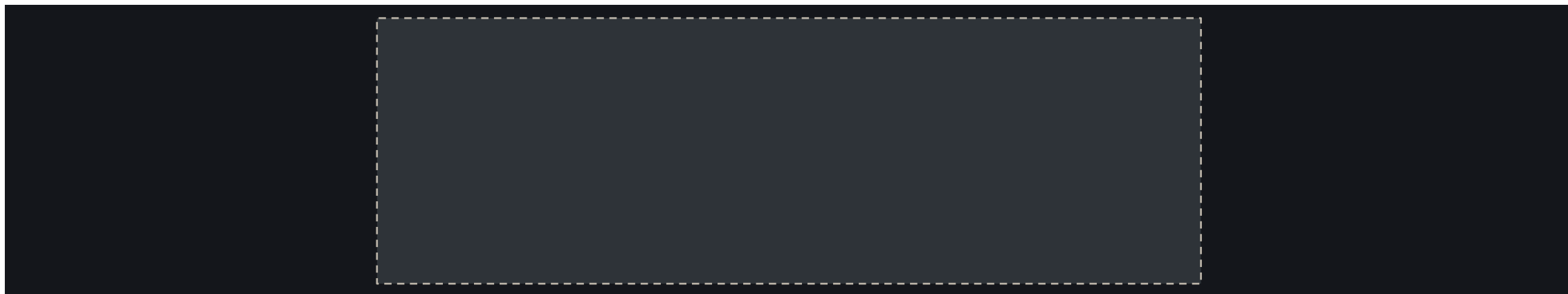
One up arrow in each box first.

PAULI EXCLUSION PRINCIPLE

The fourth arrow pairs up, pointing the opposite way.

Writing a Configuration

The same four steps, for every element.



1 Find Z

Atomic number is the electron count.

2 Fill in order

Follow the chart, lowest first.

3 Respect the caps

s holds 2, p 6, d 10, f 14.

4 Add it up

The raised numbers must total Z.

Example: Oxygen

Copy the setup, not just the answer.

PROBLEM

Write the longhand configuration and the orbital diagram for oxygen.

GIVEN

Oxygen, $Z = 8$
Neutral atom

NEED

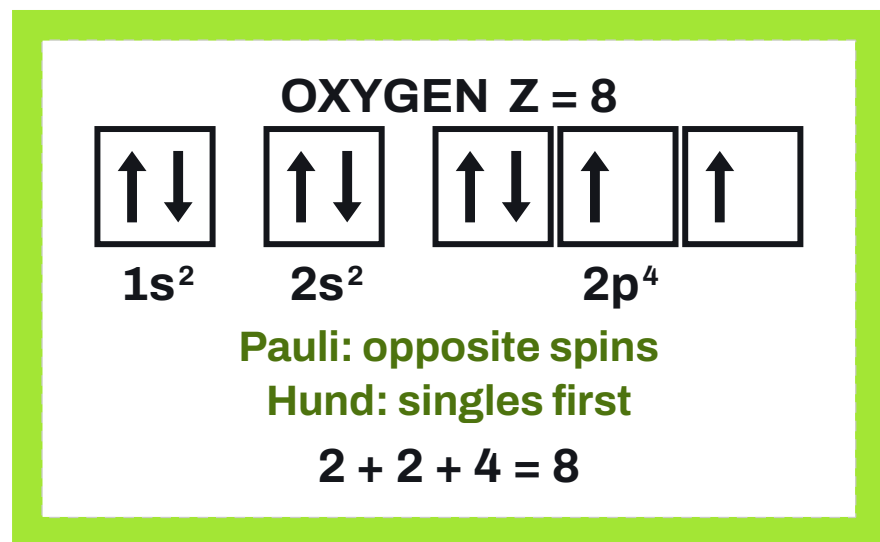
Longhand configuration
Orbital diagram
Unpaired electrons

WORK AREA**MUST WRITE**

Always check that the raised numbers add up to Z .

Oxygen, Worked

Order first. Then singles before pairs in the 2p boxes.



SETUP

Z = 8 electrons

Fill 1s, then 2s, then 2p

WORK

$1s^2 2s^2 2p^4$

$2 + 2 + 4 = 8$

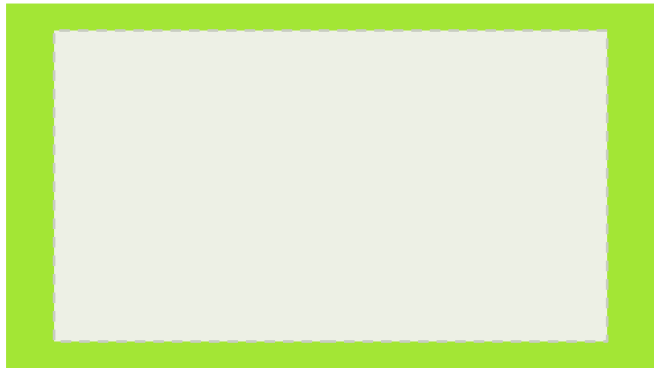
Unpaired: 2

MUST WRITE

Oxygen is $1s^2 2s^2 2p^4$, with 2 unpaired electrons.

Longhand: N, Cl, and Ca

Calcium stops at $4s^2$. The 3d sublevel stays empty until scandium.



NITROGEN, Z = 7

$1s^2 2s^2 2p^3$

Sum: $2+2+3 = 7$

Unpaired: 3

CHLORINE, Z = 17

$1s^2 2s^2 2p^6 3s^2 3p^5$

Sum: $2+2+6+2+5 = 17$

Unpaired: 1

CALCIUM, Z = 20

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

Sum: $2+2+6+2+6+2 = 20$

Unpaired: 0

MUST WRITE

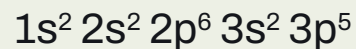
The raised numbers add up to the atomic number, every time.

Noble-Gas Shorthand

Replace the filled inner core with a noble gas in brackets.



LONGHAND: Cl



All 17 electrons written out.

SHORTHAND: Cl



[Ne] stands for the first 10: $1s^2 2s^2 2p^6$.

MUST WRITE

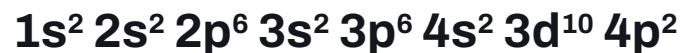
Use the previous period's noble gas. Cl is in period 3, so its core is [Ne], not [Ar].

Shorthand: Germanium

Write it long first. Then swap the argon part for [Ar].

GERMANIUM Z = 32

Longhand



Shorthand



[Ar] stands for the first 18 electrons: the noble gas before germanium

$$18 + 2 + 10 + 2 = 32 \text{ electrons}$$

STEP 1: LONGHAND

All 32 electrons, in filling order.

STEP 2: THE SUBSTITUTE

[Ar] is a placeholder for argon's 18 electrons.

STEP 3: FINISH

Then write what is left, the usual way.

Why [Ar] and Not [Kr]?

Germanium is in period 4. Its core is the noble gas that ends period 3.

	1-2	3-12	13	14	15	16	17	18
3	Na–Mg	—	Al	Si	P	S	Cl	Ar
4	K–Ca	Sc–Zn	Ga	Ge	As	Se	Br	Kr

✓ **Ar: ends period 3, BEFORE Ge**

✗ **Kr: ends period 4, AFTER Ge**

Ge: period 4, group 14

ARGON, Z = 18

Ends period 3. It comes BEFORE germanium.

GERMANIUM, Z = 32

Period 4, group 14. Use the previous period's noble gas.

KRYPTON, Z = 36

Ends period 4. It comes AFTER germanium, so it is the wrong core.

Example: Iron

Longhand, shorthand, and the unpaired count.

PROBLEM

Write longhand and shorthand for iron. How many unpaired electrons?

GIVEN

Iron, $Z = 26$
Neutral atom

NEED

Longhand
Shorthand
Unpaired electrons

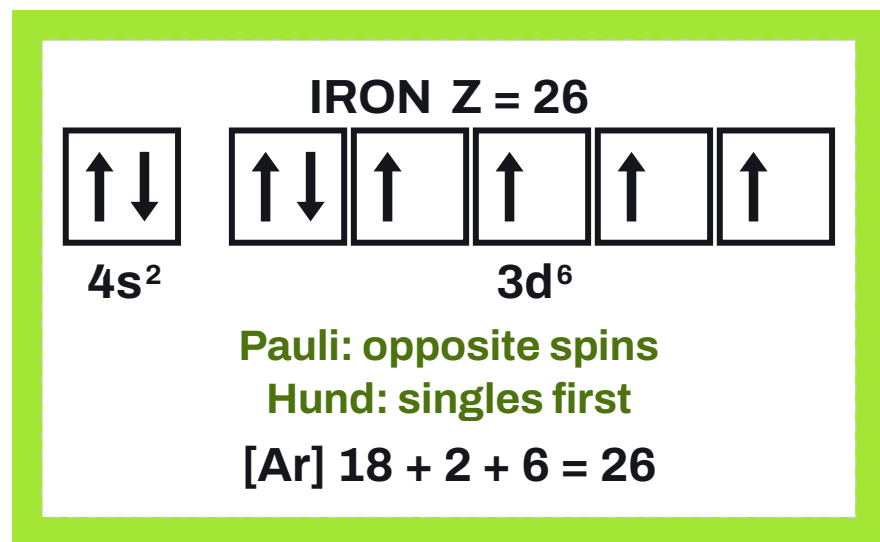
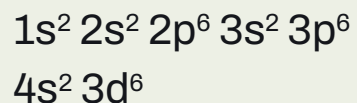
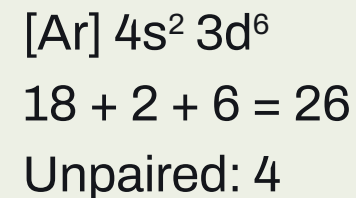
WORK AREA

MUST WRITE

Fill 4s before 3d. We write 4s ahead of 3d.

Iron, Worked

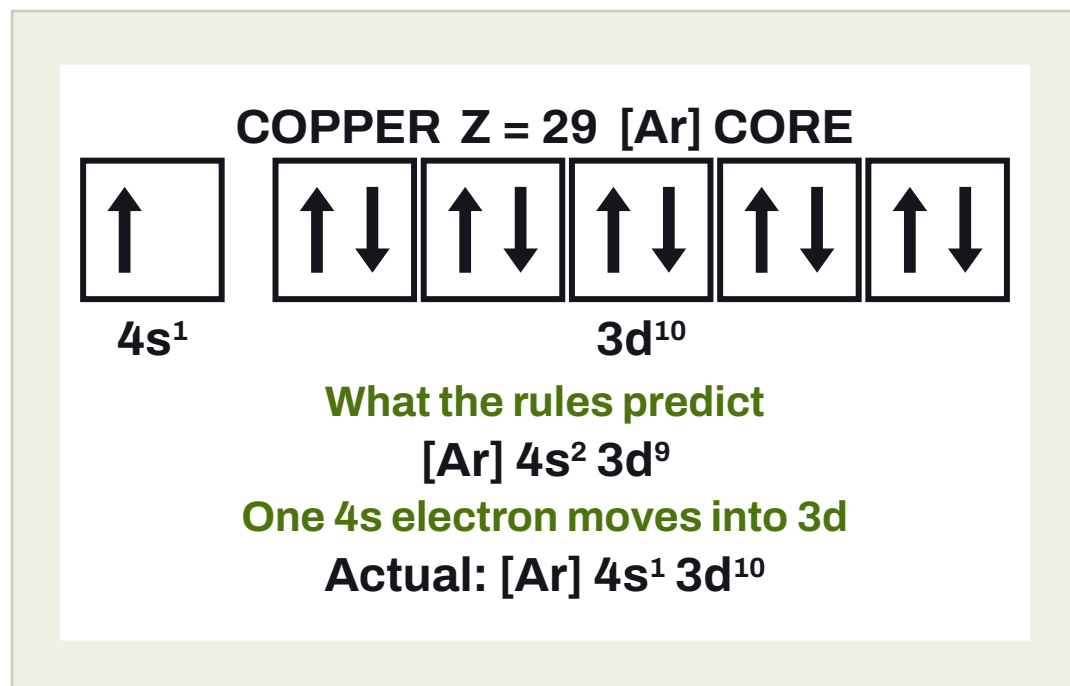
Argon is the core. Then 4s, then 3d.

**LONGHAND****WORK****MUST WRITE**

Fe is [Ar] 4s² 3d⁶. One pair and 4 singles in the 3d boxes.

Exception: Copper

The rules predict $4s^2 3d^9$. Watch what the real atom does.



WATCH

Which sublevel loses an electron, and which gains one?

ONE OF TWO

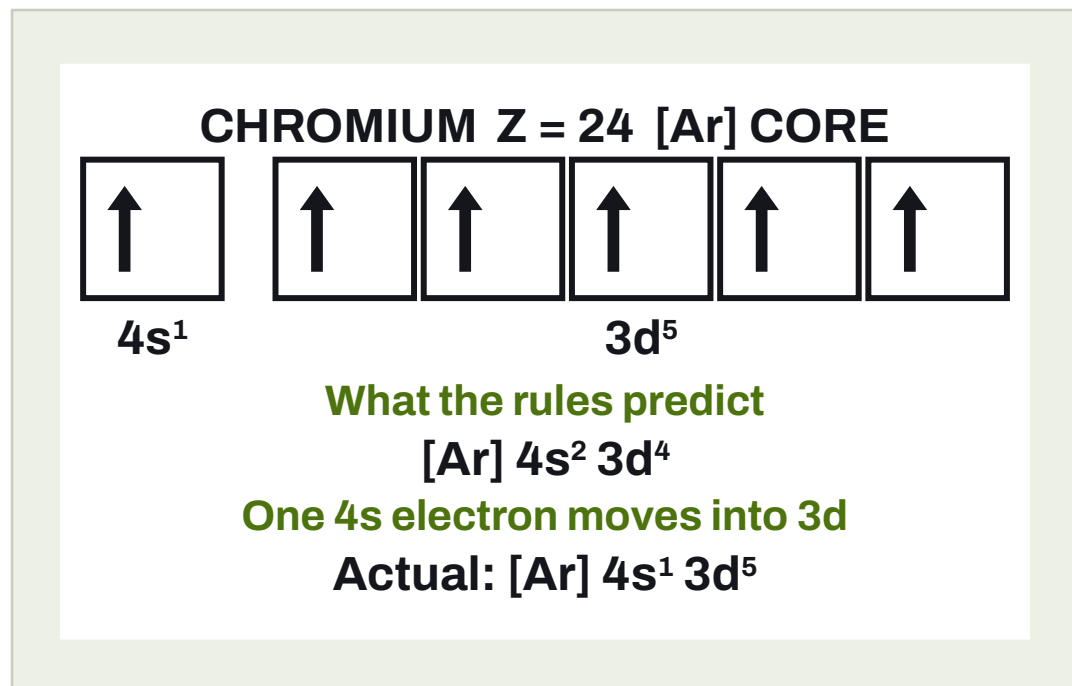
Copper and chromium are the only exceptions we learn.

LOOKING AHEAD (S2.4)

4s empties first in ions: Cu⁺ is [Ar] 3d¹⁰, Cu²⁺ is [Ar] 3d⁹.

Exception: Chromium

Same move. The rules predict $4s^2 3d^4$.



WATCH

Count the arrows in 3d before and after.

THE TOTAL

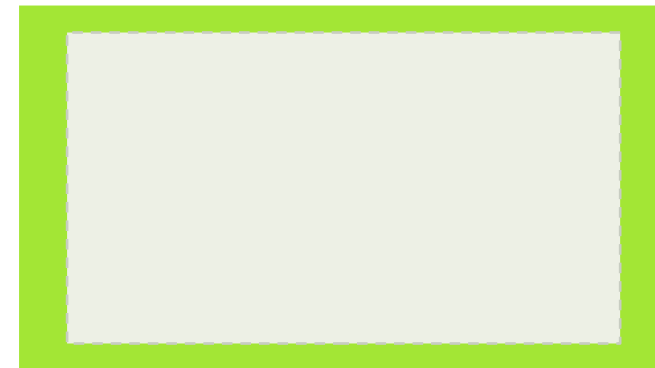
Cr still has 24 electrons. [Ar] is 18, so 6 are drawn.

UNPAIRED ELECTRONS

Count them at the end. Is it more than iron's 4?

Why Cr and Cu Break the Pattern

A simplified reason, and where it stops working.



THE ENERGIES ARE CLOSE

4s and 3d are almost the same energy, so a small change can tip the balance.

THE PAYOFF

One 4s electron moves. 3d becomes completely full (Cu) or half full with parallel spins (Cr).

THE RESULT

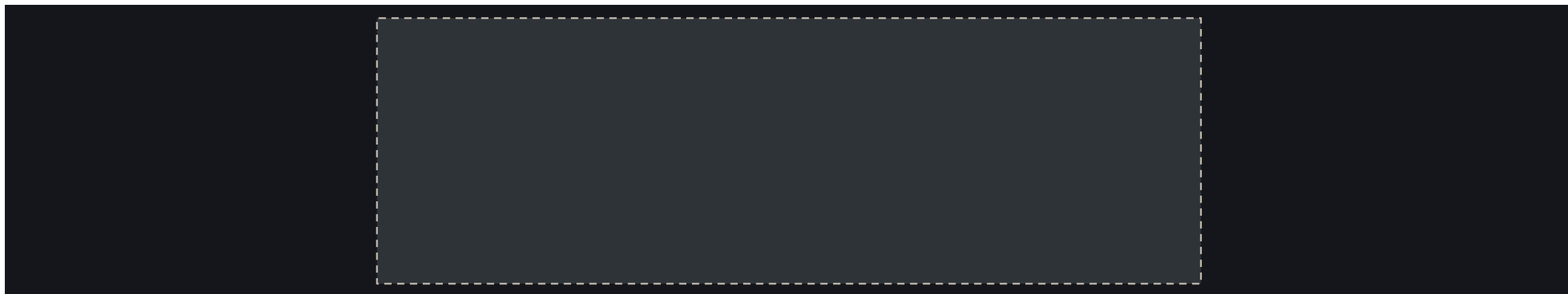
That lowers the atom's total energy slightly, so the real atom does it.

MUST WRITE

A simplification, not a rule: it misses other exceptions. Learn Cr and Cu; look the rest up.

Checking a Configuration

Run these four checks in order.



1 Count

Do the raised numbers add up to Z?

2 Caps and order

Max 2, 6, 10, 14. Aufbau order.

3 Hund and Pauli

Singles first. Pairs have opposite spins.

4 Core, Cr, Cu

Previous noble gas? Is it Cr or Cu?

Example: Spot the Mistake

Each was written as a ground state.

PROBLEM

Each of these is wrong. Name the rule or fact it breaks.

GIVEN (a), (b)

(a) O: $1s^2 2s^2 2p^6$

(b) K: $[\text{Ar}] 3d^1$

GIVEN (c), (d)

(c) Ca: $[\text{Ne}] 4s^2$

(d) Cr: $[\text{Ar}] 4s^2 3d^4$

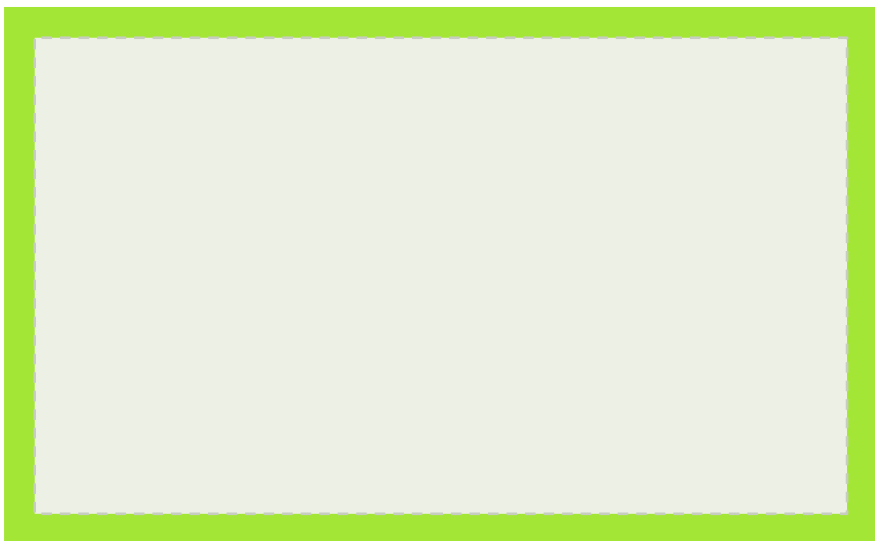
WORK AREA

MUST WRITE

Add the electrons first. If the total is not Z, stop there.

Spot the Mistake, Worked

Four problems, four different checks. Each fix is on the right.



WHAT IS WRONG

- (a) Count: 10, not 8
- (b) Aufbau: 4s first
- (c) Core: needs [Ar]
- (d) Real Cr differs

WORK

Corrected:

- (a) $1s^2 2s^2 2p^4$
- (b) $[\text{Ar}] 4s^1$
- (c) $[\text{Ar}] 4s^2$
- (d) $[\text{Ar}] 4s^1 3d^5$

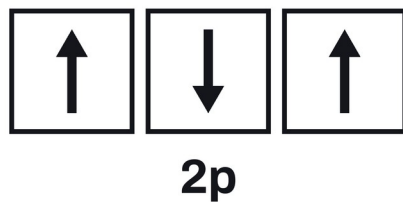
MUST WRITE

Cr follows the rules on paper, but the measured ground state is $[\text{Ar}] 4s^1 3d^5$.

Example: Spot the Mistake, Diagrams

Two orbital diagrams. Each breaks a different rule.

(e) N



(f) Ne



(E) NITROGEN, $Z = 7$

Only the three 2p boxes are drawn.

(F) NEON, $Z = 10$

Only the 2s box is drawn.

YOUR JOB

Name the broken rule. Redraw it correctly.

Spot the Mistake, Diagrams, Worked

One Hund error and one Pauli error. The fixes are drawn at left.



2p



2s

WHAT IS WRONG

(e) Hund: spins differ

(f) Pauli: same spin

WORK

Fixed:

(e) 2p: up, up, up

(f) 2s: up, down

MUST WRITE

Hund: singles first, all spins the same. Pauli: two in one orbital must be opposite.

Check Yourself

Work all three in your notes before we compare.

PHOSPHORUS, $Z = 15$

Write the longhand configuration.

SULFUR, $Z = 16$

Write the shorthand and sketch the 3p boxes.

COPPER, $Z = 29$

Write the actual shorthand. Say why it is an exception.

MUST WRITE

Show the electron sum for each one. If it is not Z , find the error.

LOOKING AHEAD

Where This Goes Next

Next: valence electrons, and the
configurations of ions.